

Introduction

- Need for cane with sensors to help Visually Impaired
 - Using a voice prompt model and haptic feedback will allow the Visually Impaired to be more independent in travel
 - It will become faster overtime with improvements in software and hardware
- Independence of Visually Impaired will be improved because they will not need an aid to help them
- A new set of eyes for Visually Impaired

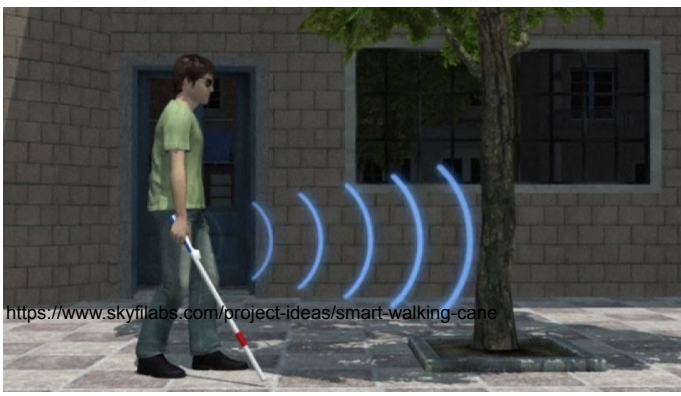


Figure 1:Visually Impaired person holding a cane that uses sonar to detect if objects are in front of the Visually Impaired

Vocab:

- MOT: Multiple Object Tracker
- AI: Artificial Intelligence

Review of Literature

Object Tracking Using Computer Vision: A Review

Kadam, P., Fang, G., & Zou, J. J. (2024). Object Tracking Using Computer Vision: A Review. Computers, 13(6), 136. <https://doi.org/10.3390/computers13060136>

- Goal: To provide an in-depth overview of object tracking in computer vision by exploring different sensors and methods, along with the pros and cons of each approach.
- Key Findings: Deep learning has significantly advanced object tracking, and creating a generalized framework for tracking and detection could greatly enhance accuracy and efficiency in robotics and autonomous vehicles.

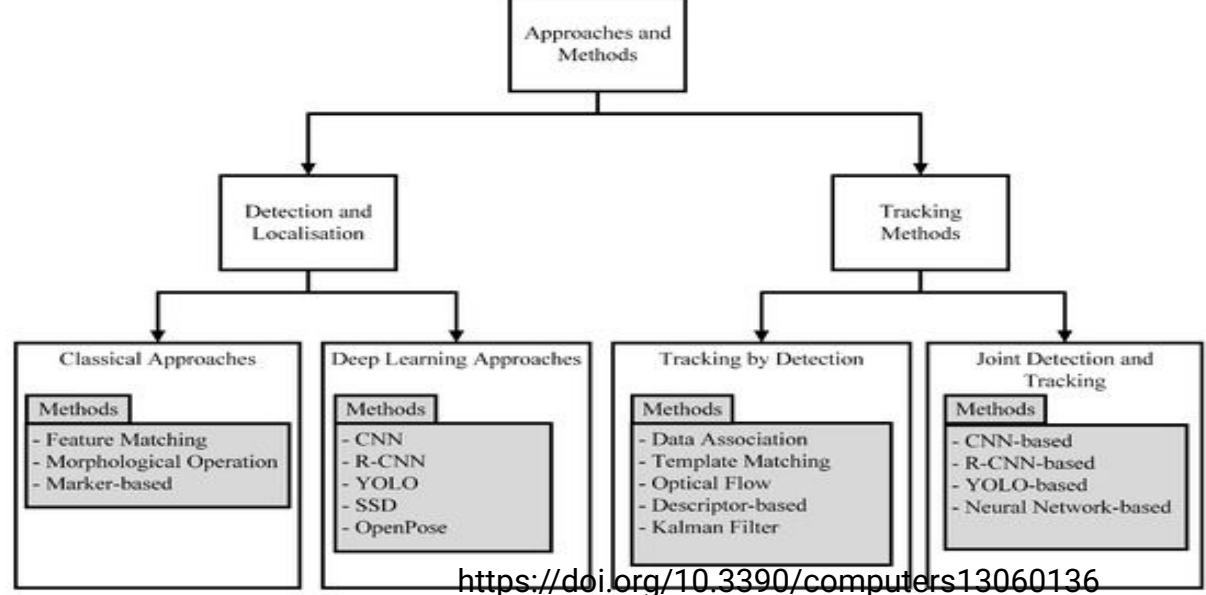


Figure 2:variations of approaches and methods for object tracking

DefectTrack: Multi-Object Multi-Camera Tracking Based on Deep Learning for Intelligent Transportation

Fei L, Han B. Multi-Object Multi-Camera Tracking Based on Deep Learning for Intelligent Transportation: A Review. Sensors (Basel). 2023 Apr 10;23(8):3852. doi: 10.3390/s23083852. PMID: 37112193; PMCID: PMC10144185.

- Goal:This article reviews multi-object multi-camera tracking methods, outlining how to implement them, benchmark system performance, and identify common challenges in the field.
- Key Findings: Deep learning methods like DM-MOT and GN-MOT improve tracking accuracy by enhancing object association across frames; DM-MOT uses Siamese networks and Convolutional Neural Networks (CNNs) to learn similarity models and track objects based on distance.

CNN-Based Object Recognition and Tracking System to Assist Visually Impaired People

3. F. Ashiq et al., "CNN-Based Object Recognition and Tracking System to Assist Visually Impaired People," in IEEE Access, vol. 10, pp. 14819-14834, 2022, doi: 10.1109/ACCESS.2022.3148036.

- Goal: To develop a smart assistive system for Visually Impaired individuals that uses deep learning for real-time object detection and includes a web-based interface for family members to monitor their safety.
- Key Findings: Comparing consecutive video frames using the Structural Similarity Index (SSIM) improves efficiency by skipping identical frames, reducing computational load during real-time object tracking.

Problem Statement

- The reviewed articles conclude that real-time object tracking faces significant challenges, especially when adapted for mobile use.
- Innovations are needed to make these systems effective and practical for assisting Visually Impaired individuals in dynamic environments.
- The research focuses on developing affordable, modern visual aids to replace outdated tools—some of which have seen little innovation since the 1800s.
- The goal is to create accessible technology that enhances mobility and independence for all Visually Impaired users.

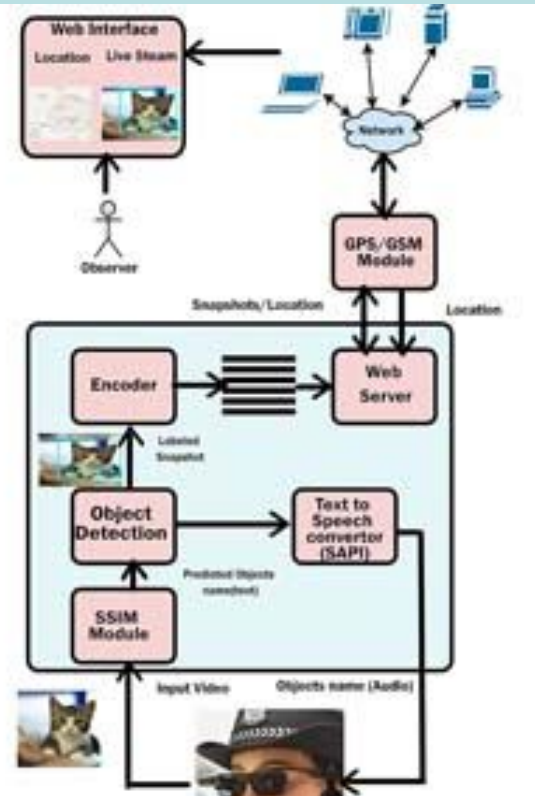


Figure 3: Object recognition to help Visually Impaired structures laid out.

Goals

- This proposed research will combine AI and something not innovated on for centuries to make Visually Impaired people independent and able to contribute to society without the need of a human aid which they want.
- Combining a Cane and an AI program that can detect objects and sidewalks and give directions to the Visually Impaired.
- It should work because previous studies have been done but never made it to market and the only attempt at this idea is a cane called we walk for 8 grand that does something but doesn't have all the functionality needed.
- It is plausible to make the program run and it should work on a laptop. The questionable part is integration onto a phone and still have it ruin fluidly.

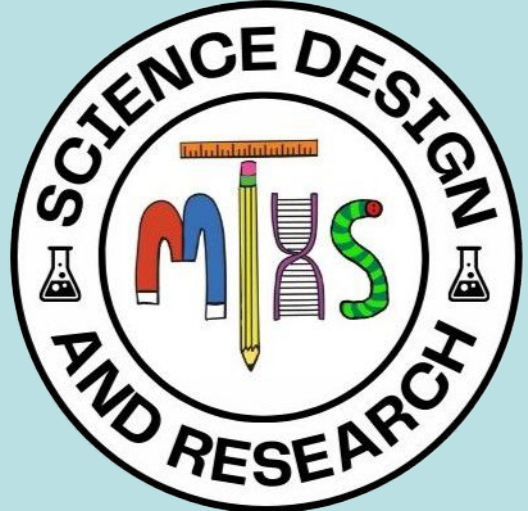
Hypothesis

- Early experiments may face challenges, but the core concept of the project—using object tracking to aid the Visually Impaired—will be positively received.
- Developing AI systems like object tracking involves time-intensive training and trial and error, but research shows that enhancing independence is a key need for Visually Impaired individuals, making this solution both relevant and valuable.

Object Tracking With AI To Help The Visually

Impaired

Logan Fitzpatrick



Methodology

Step 1

- Construct an object tracking system with distance integration
 - It would use yolo v4 nano and testing would happen in the same stage
 - It needs to be able to make it run on a raspberry pi so it is easier to run on a phone later
 - As long as 30 fps is possible the object tracker is possible

Step 2

- Develop a sidewalk and curb detection program separately
 - Testing has to be done independently.This is going to be a more taxing program so hopefully the results will show an fps of 15 or greater
 - The biggest objective here is to make it work on a laptop. It's easier to shrink something versus create something super tiny at the start

Step 3

- Combine programs
- The programs need to be combined to see if they can both function accurately
- This entails getting a close enough distance which would be within 8 inches, getting the correct object 80% of the time or higher
- Run it all on a laptop

Step 4:

- Develop an AI that can interpret the data and output it in a way a human can understand such as directions or audiating what the object is in bluetooth headphones
- This had to be tested to give directions which means going around something, also in different lighting and weather conditions, this has to be added to the program and also run on laptop with the trackers still working fluidly whatever that FPS may be

Step 5

- Shrink the program and get it to run on a brand new iphone or samsung phone. Iphone would be better because it's usually less powerful then samsung for RAM intensive tasks which AI is
 - Cut down on objects known, limit weather curb detection versus sidewalk detection is enough or more important
 - See if with the current technology out their if it's even possible to run on a phone or maybe does the program have to run on an ipad or tablet
 - The biggest thing is figuring out what portable device that's light weight can run the air on the cane
 - It also has to keep the cane balanced

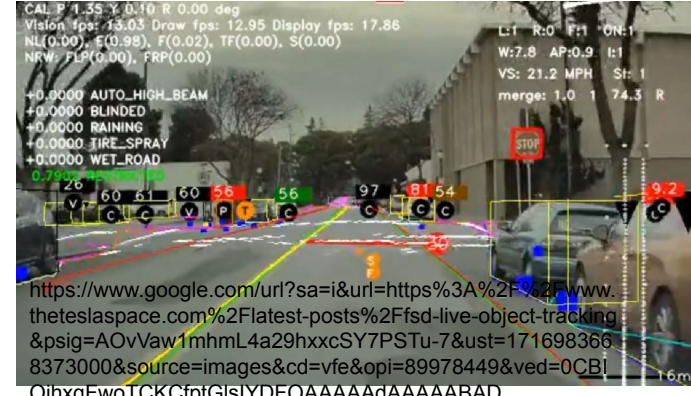


Figure 4: Self driving with MOT tracking



Figure 5:Tesla Autopilot - What the Car sees

Variables

- Control: Each individual program functioning independently without integration of additional components.
- Variables
- Weight of the smart cane and its impact on usability.

- Number of people the system can accurately track at once.
- Accuracy of each program operating on its own versus when combined with other programs.

Results

Graph 1: Track Lifespan

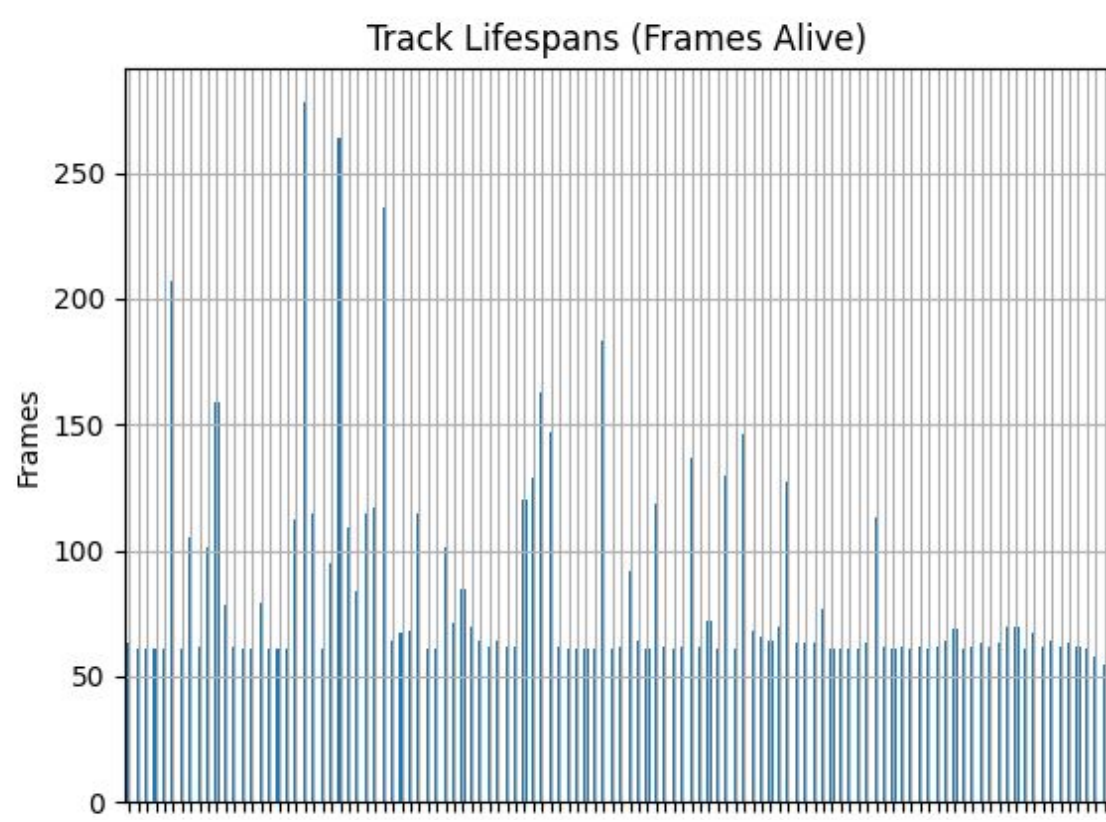


Figure 6:This bar chart illustrates how long each unique object track persisted, indicating variability in tracker performance across different objects.

Graph 2: Tracking Stability

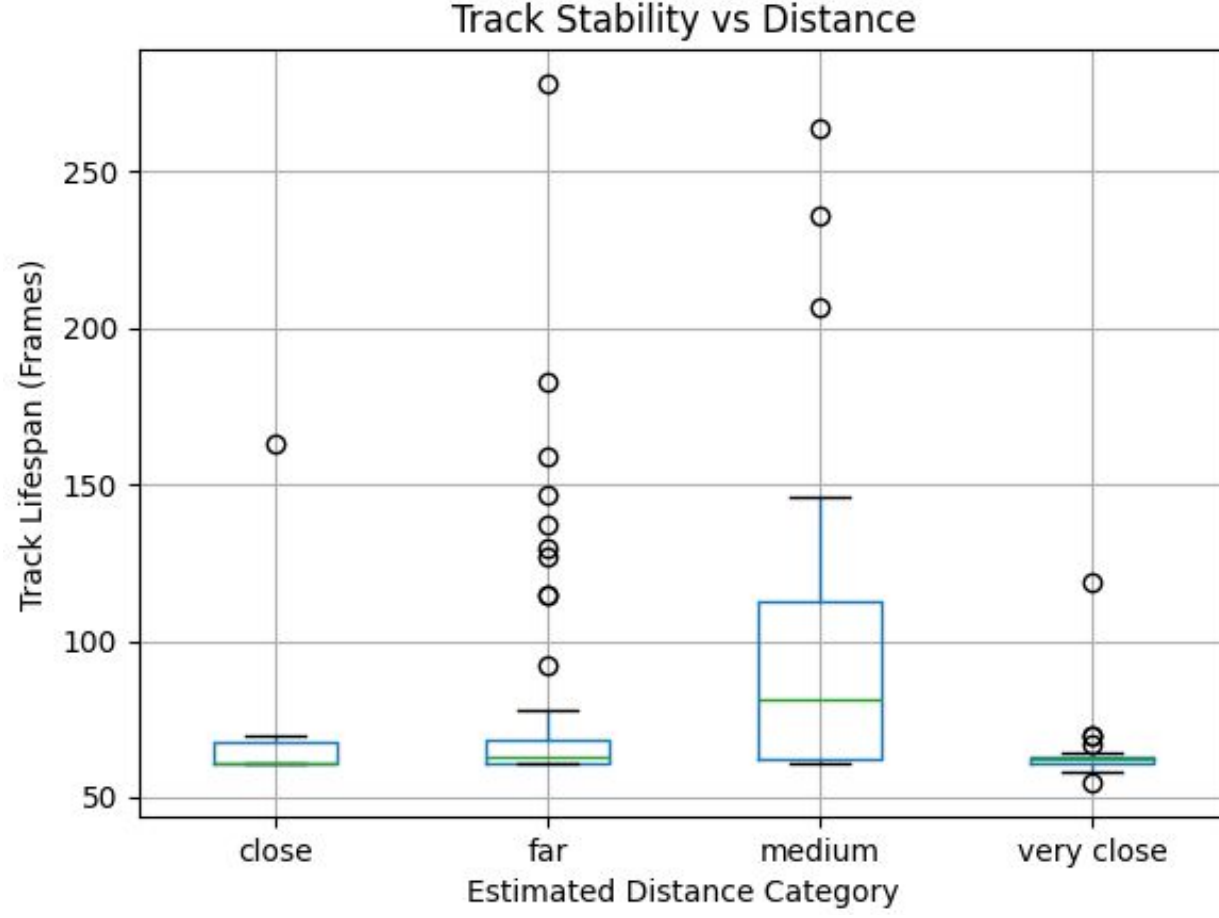


Figure 7: This box plot compares track lifespans (in frames) across estimated distance categories, showing how distance affects tracking reliability.

Graph 3: Average Bounding Box Size

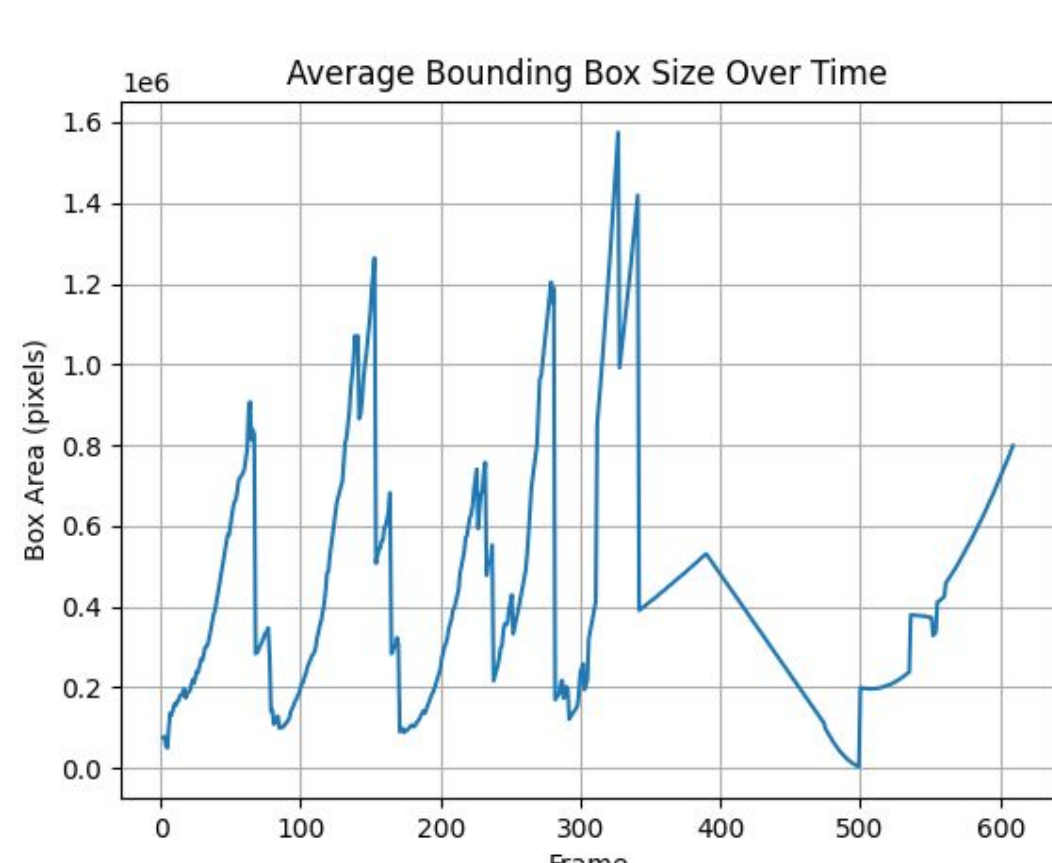


Figure 8: This line chart shows the average size of detected bounding boxes across video frames

Graph 4: People Tracked

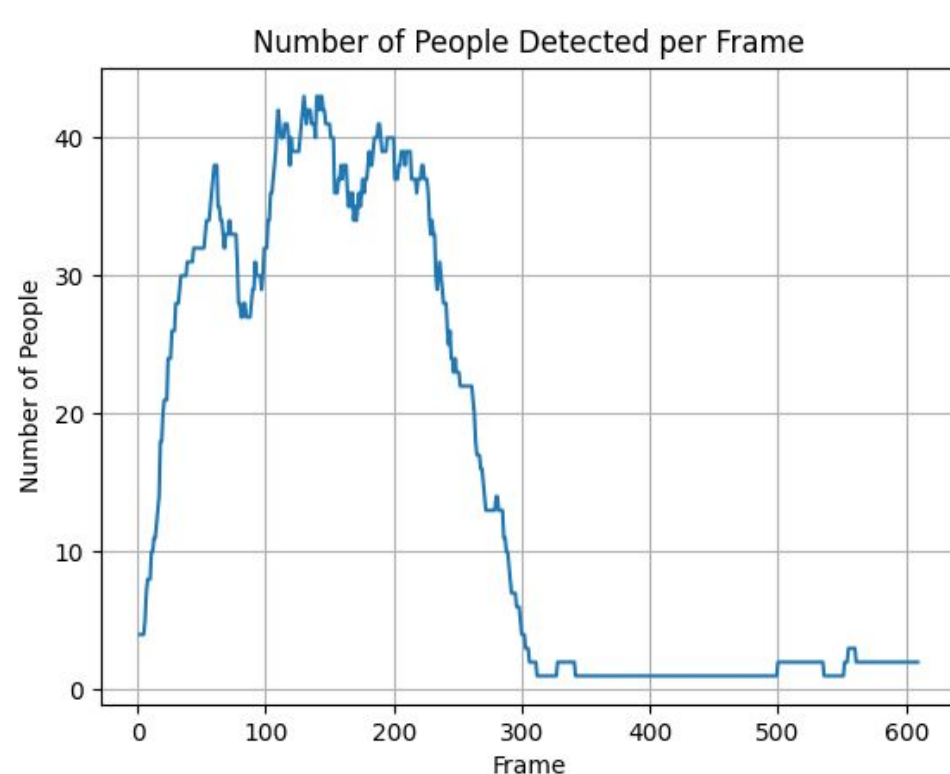


Figure 9: This time series tracks how many individuals are detected in each frame, revealing crowd density and detection consistency over time.

Graph 5: Tracking Confidence

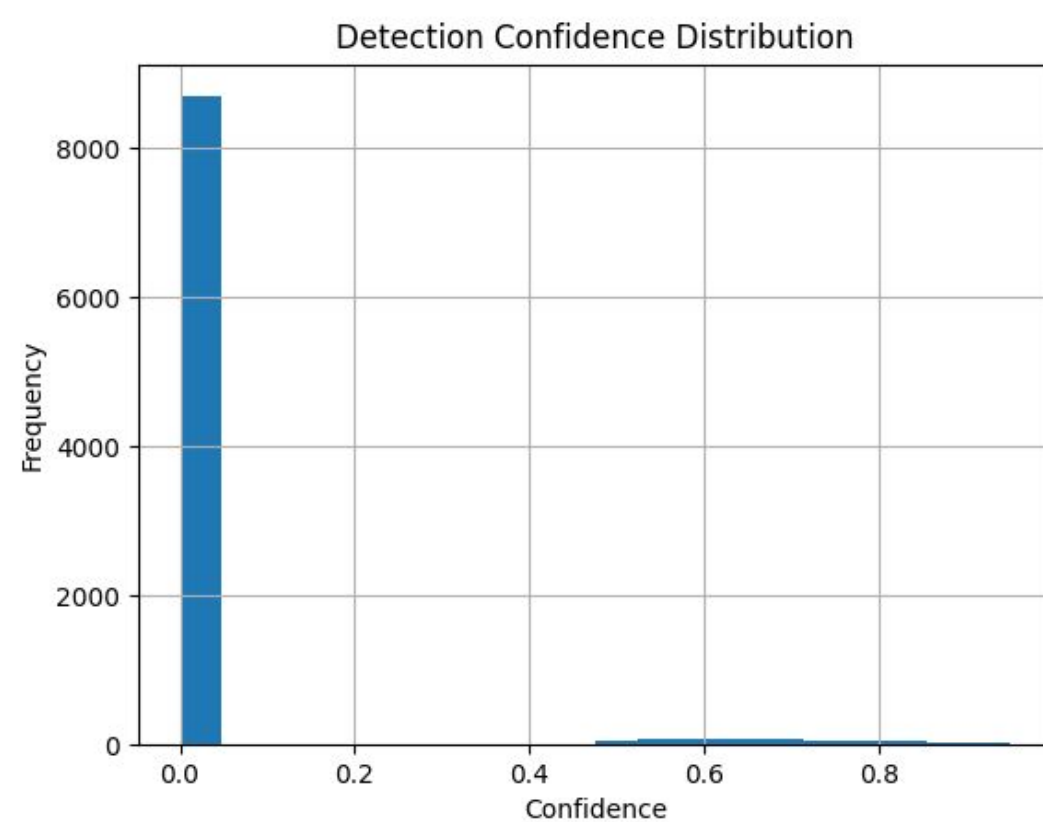


Figure 10: This histogram visualizes how frequently different detection confidence scores occur, with the majority at low values.

Based on the results we can draw some conclusions about the tracking and usefulness of the tracker

Analysis & Discussion

All these graphs are important because they all show an analysis for evaluating the object detection and tracking performance in a real world application.

- *Bounding Box Size Over Time* shows how object size varies, helping diagnose zooming, occlusions, or perspective issues.
- *Confidence Distribution* highlights model reliability—low scores may signal poor detections or challenging video frames.
- Crowd Density and Scene Complexity
 - *People Detected per Frame* reveals how busy the scene is and whether detection drops occur during occlusions or lighting shifts.
 - It shows that in city situations the detection doesn't work the best.
- Tracking Stability Insights
 - *Track Stability vs Distance* helps evaluate how well the tracker performs at varying depths, with longer lifespans shows better tracking.
 - Useful for tuning trackers in environments with changing camera angles or movement patterns.
- Overall Tracker Performance
 - *Track Lifespans (Frames Alive)* reflects consistency in object tracking—shorter tracks may indicate fragmentation, re-identification failures, or lost objects.

Importance & Application

- Allows Visually Impaired individuals to navigate environments without constant human assistance.
- Can be used in other applications such as an evaluator for guide dogs once it reaches that capacity because a program will be consistent.
- Reduces the risk of accidents by providing situational awareness through audio or haptic feedback.
- Learns from the user's environment and behavior, improving accuracy and usability over time.
- Can connect with smartphones or cloud systems for location tracking, emergency alerts, or caregiver monitoring.
- Offers an affordable, scalable alternative to more expensive assistive devices or guide animals.

Conclusion

- Problems:
 - New field so minimal research.
 - Currently very expensive to implement.
 - Have to use large computers and servers over the air to use the program so it can't be run locally so it's easier to get hacked.
- Goals:
 - Using a camera on a cane to help Visually Impaired with navigation and make them live more independent lives.
- Findings:
 - Making an affordable Smart Walking Cane is hard and not done yet.
 - Most MOT tracking models can detect multiple objects but have NOT trackers detect clusters of objects is extremely hard.
 - Having an AI roughly estimate how many objects are in front of it is hard because it skips frames and moves from one object to another.

Intended Future Research

- Give Visually Impaired more functional lives.
- Have Visually Impaired use this device.
- Improve this to work in cities with rapidly changing environments.
- Make a better system to be able to alert the Visually Impaired of location and direction to go.
- Use multiple sensors in the cane to make it more functional compared to only a camera currently.
- This tech could possibly be applied to glasses which is a more visually appealing application.